



Shahjalal University of Science and Technology, Sylhet.

Lab Report

Department of Electrical and Electronic Engineering

**“Design and analysis of Monopole and Dipole Antenna
in CST Studio.”**

Course Code: **EEE 476**

Course Title: **RF Lab**

Experiment No.: **06**

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Introduction

Antennas play a vital role in all wireless communication systems, serving as the key components responsible for transmitting and receiving electromagnetic waves. Among the most fundamental and widely used types are the monopole and dipole antennas—both linearly polarized and resonant structures that are simple to design and fabricate. This experiment focuses on the design, simulation, and analysis of monopole and dipole antennas resonating at 2.42 GHz using CST Studio Suite. The primary objective is to observe how antenna length influences the resonance frequency and to validate the design parameters through parametric sweep analysis.

Design Objectives

- Design and simulate a half-wave dipole antenna and a monopole antenna tuned to 2.42 GHz.
- Perform a parametric sweep for the dipole antenna to determine the optimal resonant length.
- To find radiated power and Farfield radiation for both antennas.

Theory

Dipole Antenna

A half-wave dipole antenna consists of two equal conductive arms, each roughly $\lambda/4$ in length. The total dipole length can be expressed as:

$$L_{\text{dipole}} = \lambda/2 = c / (2f\sqrt{\epsilon_{\text{eff}}})$$

For free space ($\epsilon_{\text{eff}} \approx 1$) and $f = 2.42$ GHz:

$$L_{\text{dipole}} = 3 \times 10^8 / (2 \times 2.42 \times 10^9) \approx 62.0 \text{ mm}.$$

However, due to effects such as conductor diameter and end correction, the actual resonant length is often around 0.47λ , approximately 58–60 mm. In the CST simulation, the resonance occurred at a total dipole length of 73 mm, which indicates an increase in the effective wavelength resulting from substrate and environmental influences such as dielectric loading and feed structure.

Monopole Antenna

An antenna operates above a conductive ground plane, effectively acting as half of a dipole, with the ground plane providing a mirror image. The theoretical length is given by:

$$L_{\text{monopole}} = \lambda/4 = 3 \times 10^8 / (4 \times 2.42 \times 10^9) \approx 30.99 \text{ mm}.$$

This calculated value was implemented directly in CST (31 mm) and provided satisfactory resonant performance at 2.42 GHz.

Simulation Results and Analysis

Monopole Antenna : $L = 31 \text{ mm}$

Cylinder [X]

Name:

Orientation: ☐ X ☒ Y ☐ Z

Outer radius: Inner radius:

Xcenter: Zcenter:

Ymin: Ymax:

Segments:

Component:

Material:

OK Cancel Preview Help

Fig: Monopole Antenna Body Parameters

Brick [X]

Name:

Xmin: Xmax:

Ymin: Ymax:

Zmin: Zmax:

Component:

Material:

OK Cancel Preview Help

Fig: Monopole Antenna Ground Parameters

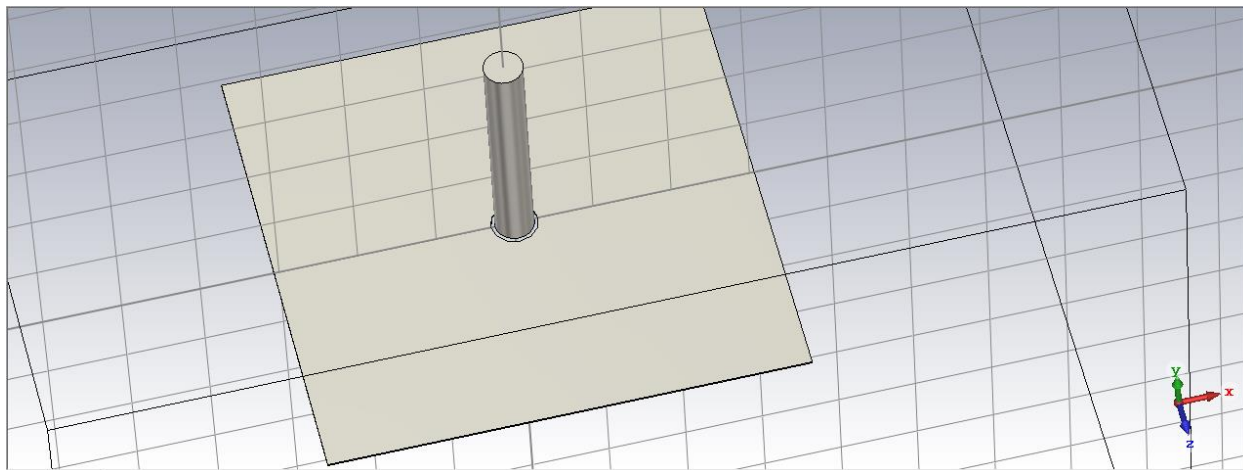


Fig: Monopole Antenna Design

Farfield Radiation Pattern-Monopole (3D):

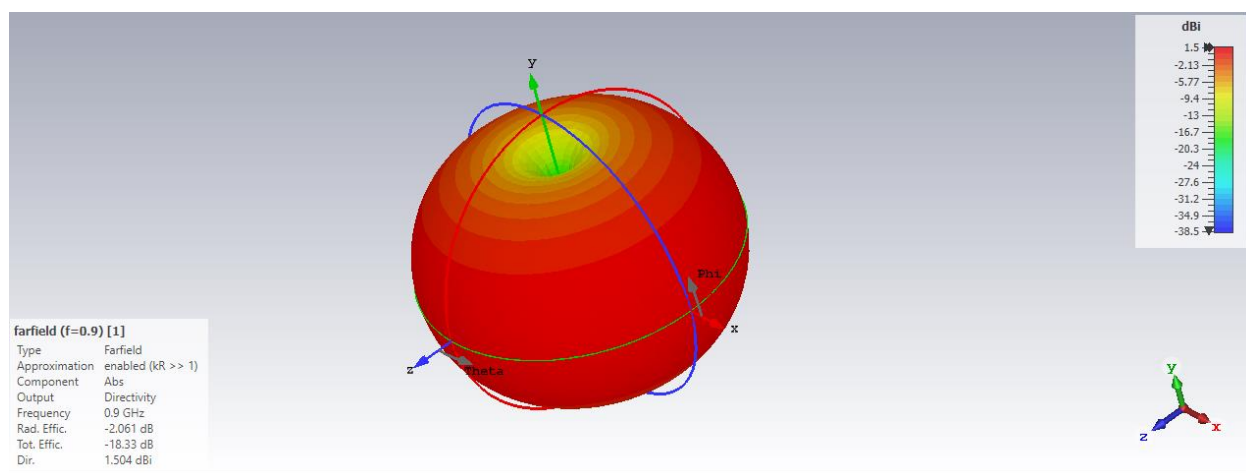


Fig: Farfield Radiation at 0.900 GHz

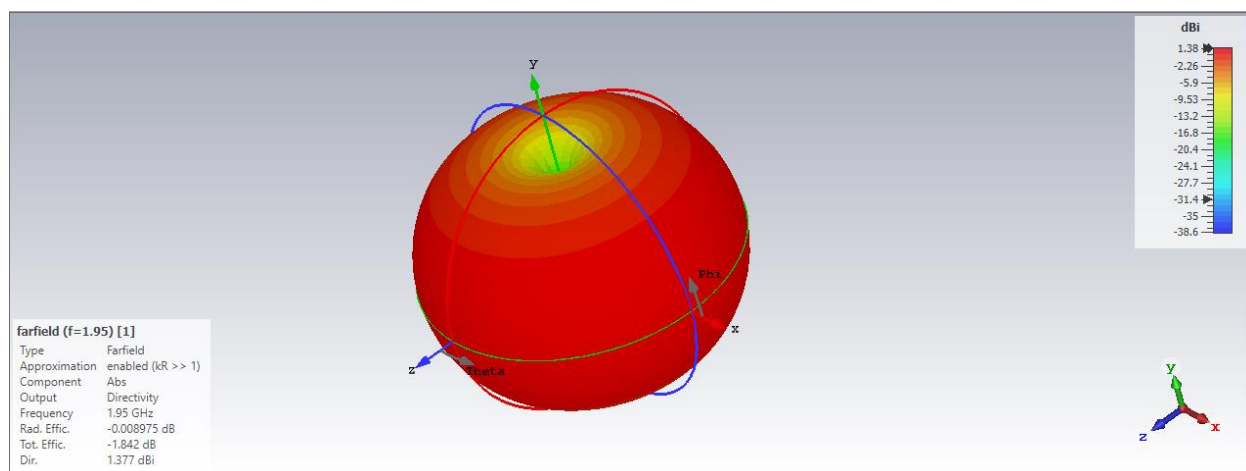


Fig: Farfield Radiation at 1.95 GHz

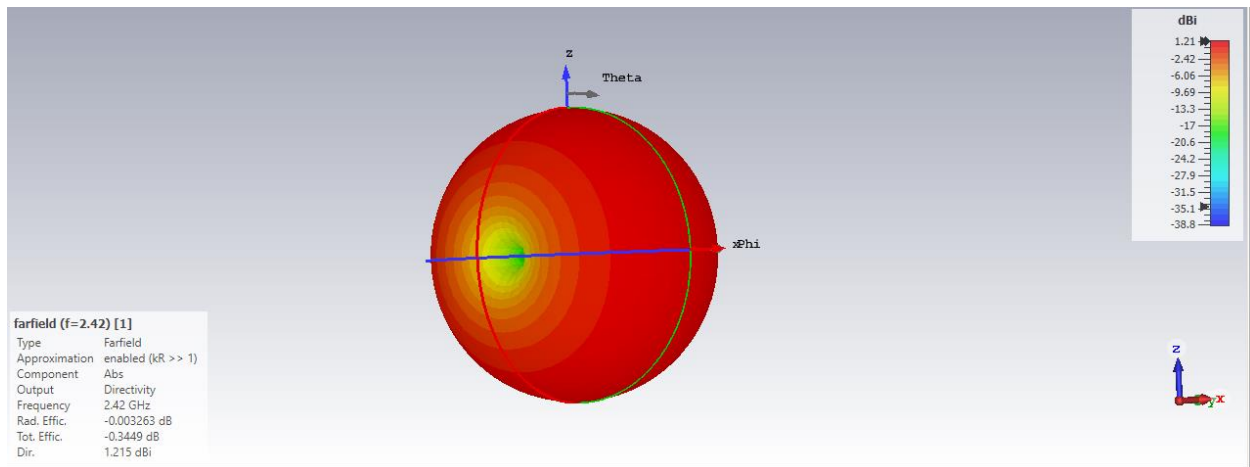


Fig: Farfield Radiation at 2.42 GHz

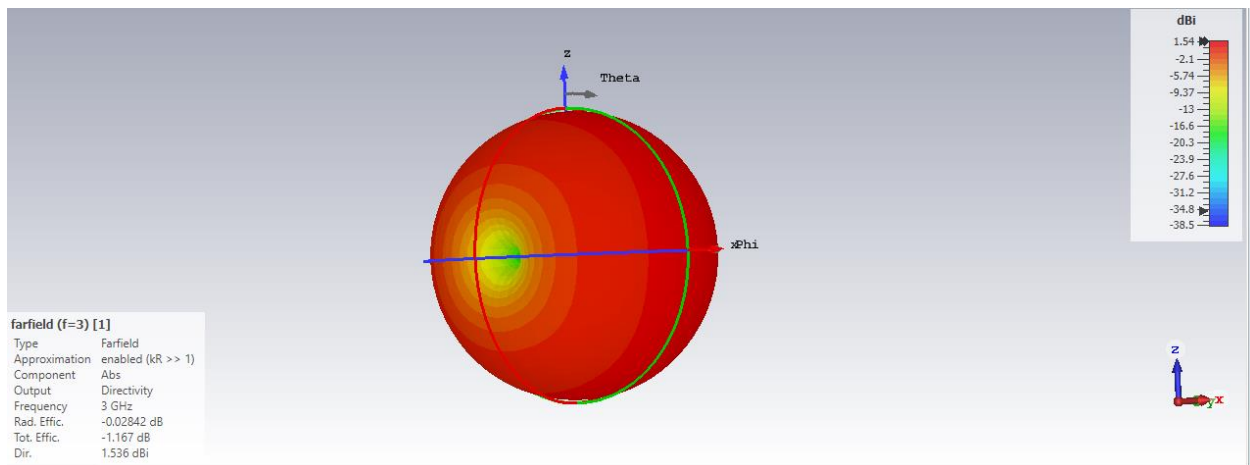


Fig: Farfield Radiation at 3 GHz

Farfield Radiation Pattern-Monopole (polar):

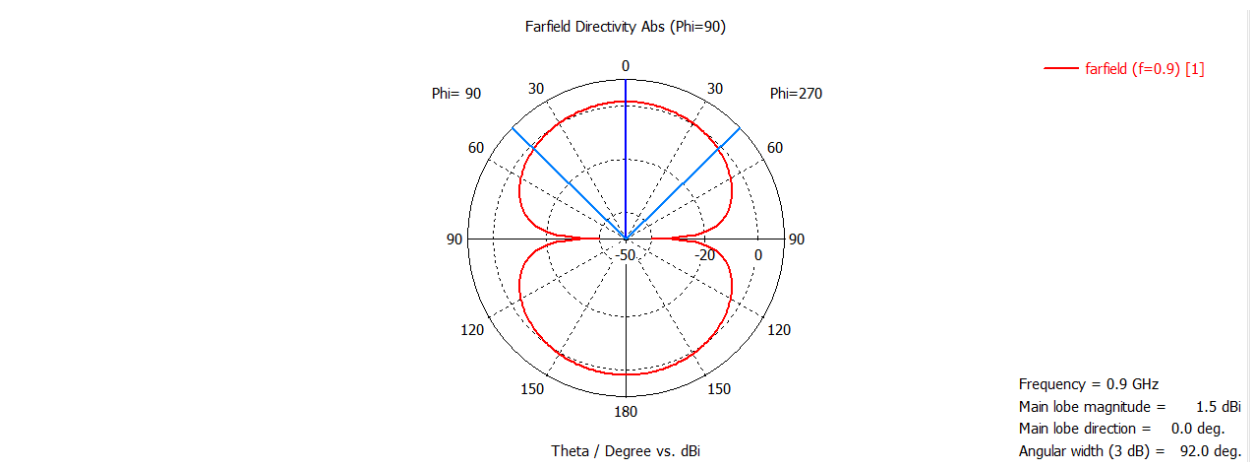


Fig: Farfield Radiation at 0.900 GHz

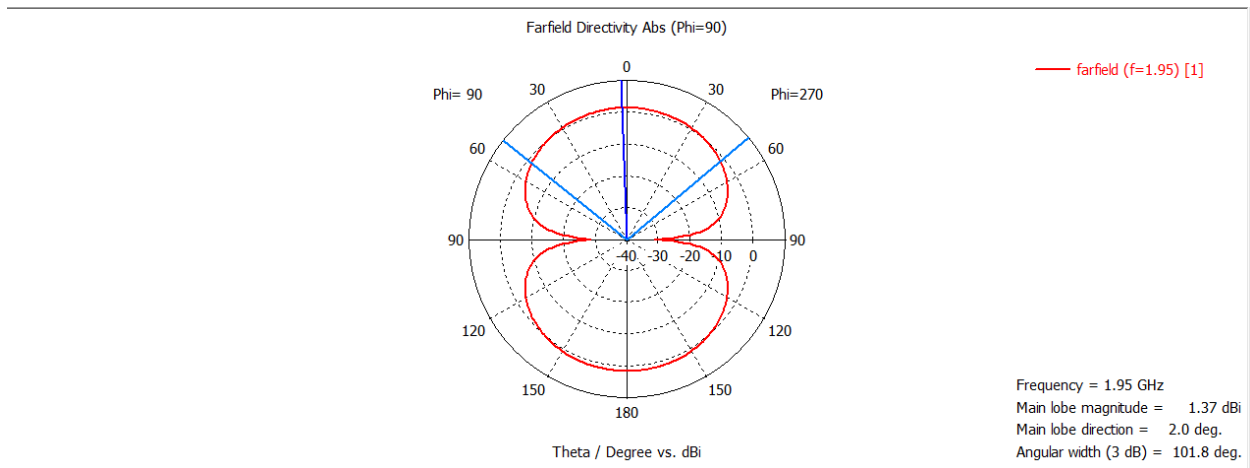


Fig: Farfield Radiation at 1.95 GHz

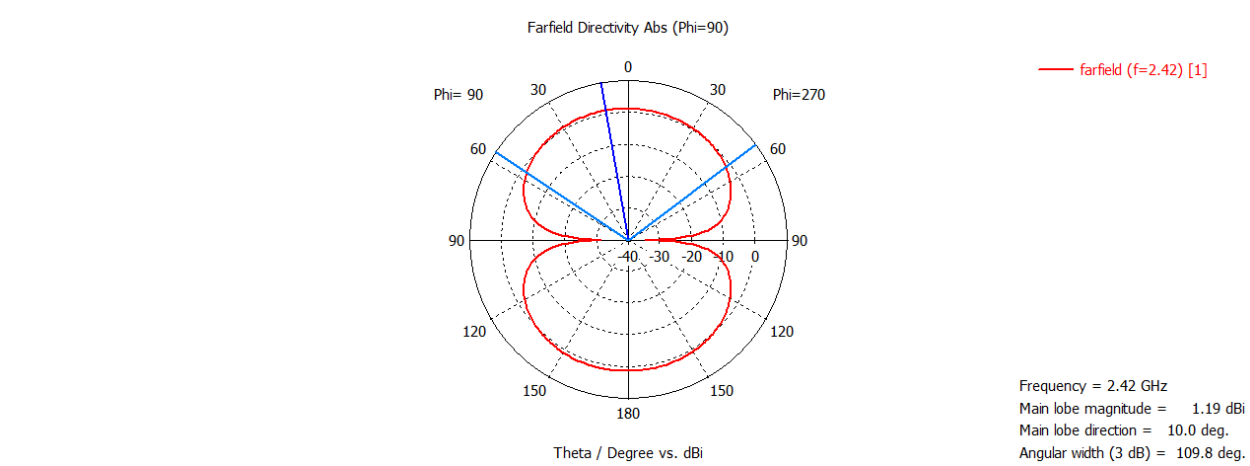


Fig: Farfield Radiation at 2.42 GHz

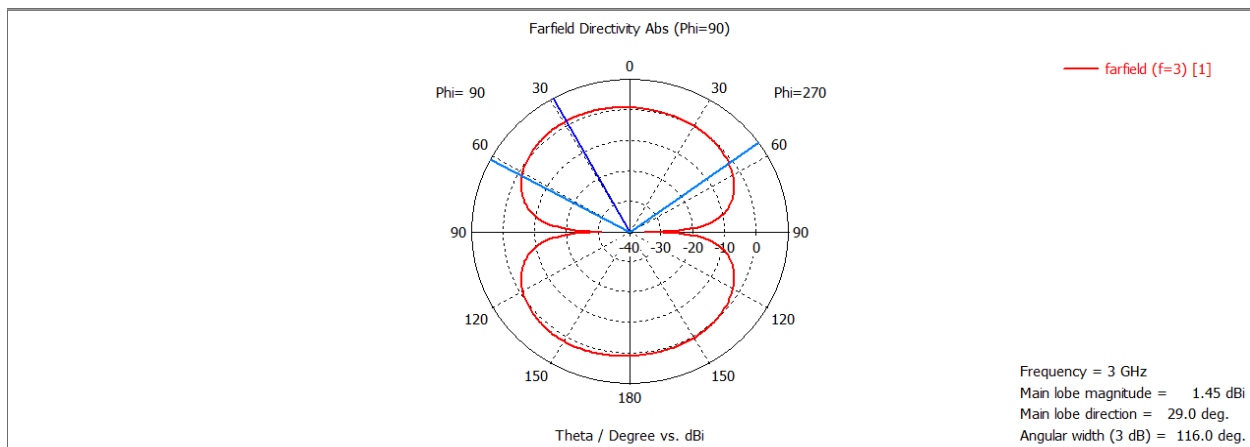


Fig: Farfield Radiation at 3 GHz

Power Radition-Monopole:

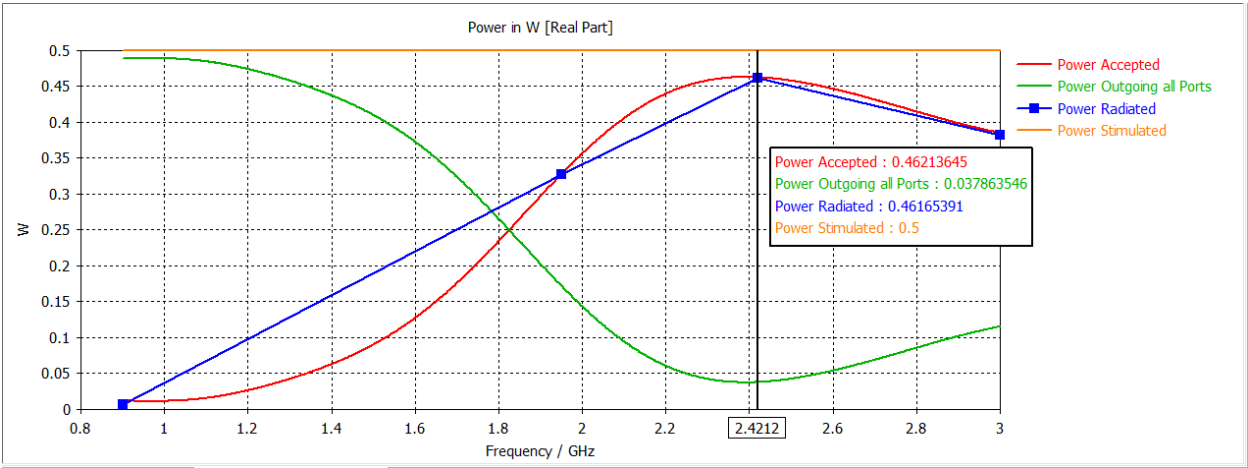


Fig: Maximum Power at Resonant Frequency 2.42 Ghz

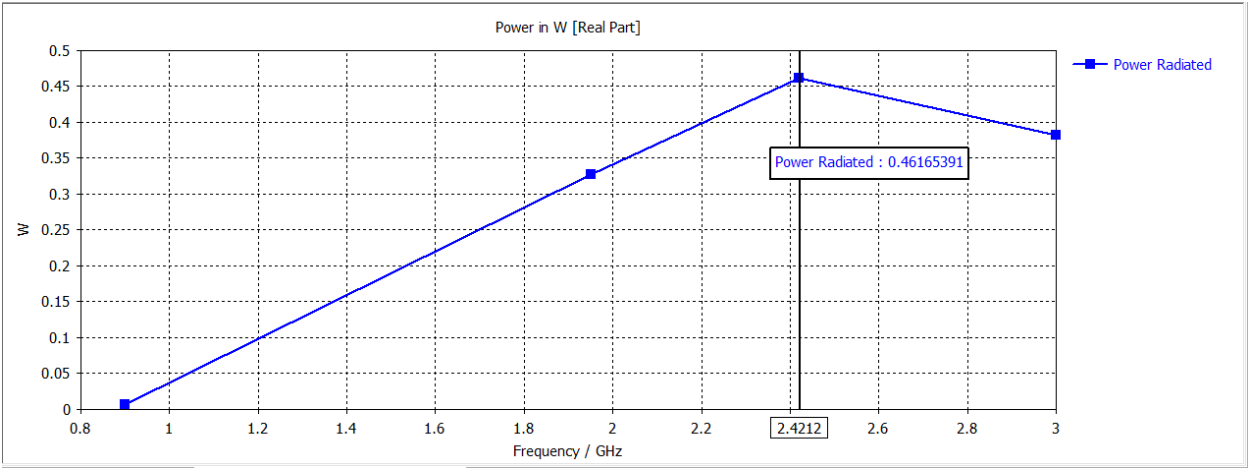
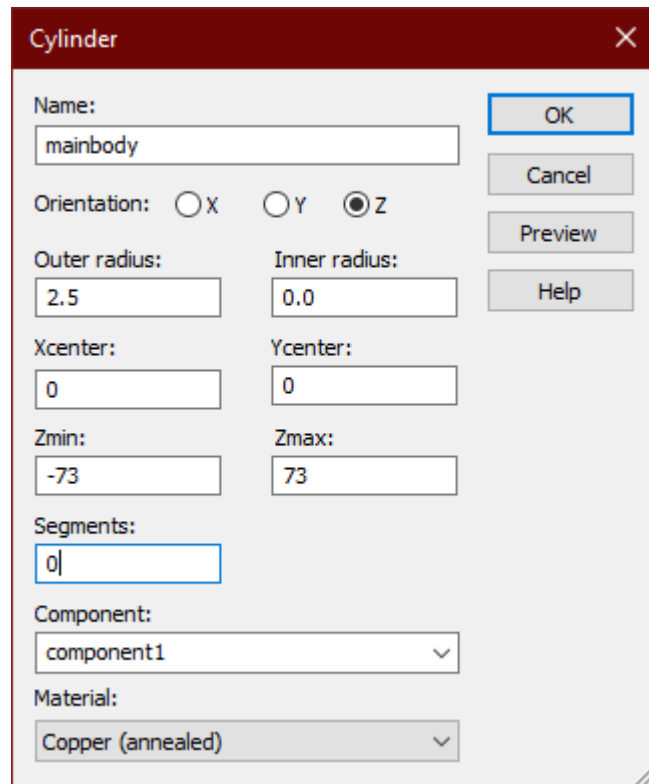


Fig: Power Radiated

Dipole Antenna: $L=73\text{mm}$



Cylinder

Name: mainbody

Orientation: ☐ X ☐ Y ☒ Z

Outer radius: 2.5 Inner radius: 0.0

Xcenter: 0 Ycenter: 0

Zmin: -73 Zmax: 73

Segments: 0

Component: component1

Material: Copper (annealed)

OK Cancel Preview Help

Fig: Dipole Antenna Body Parameters

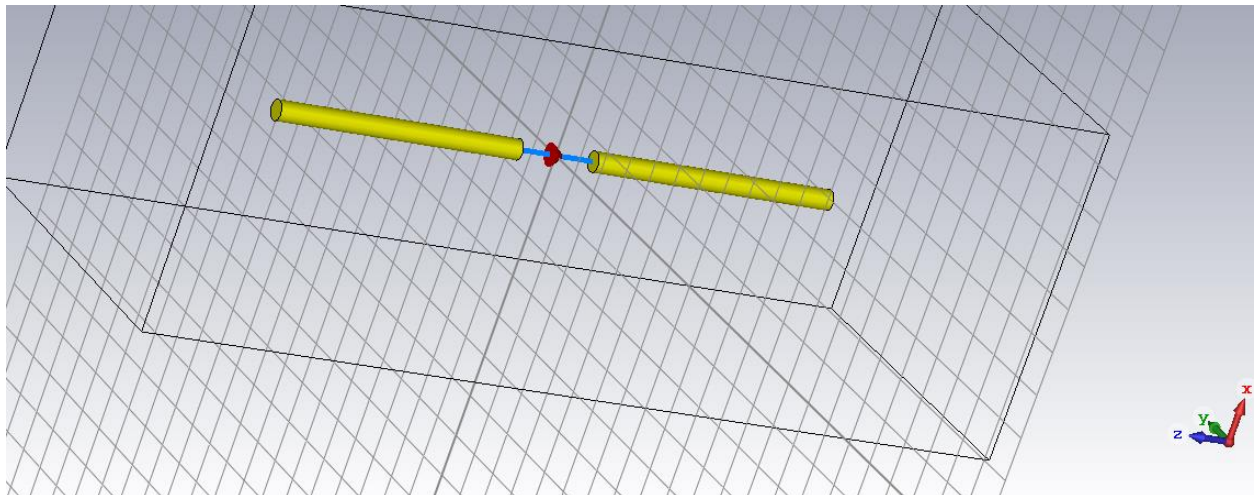


Fig: Dipole Antenna Design

Farfield Radiation Pattern-Dipole (3D):

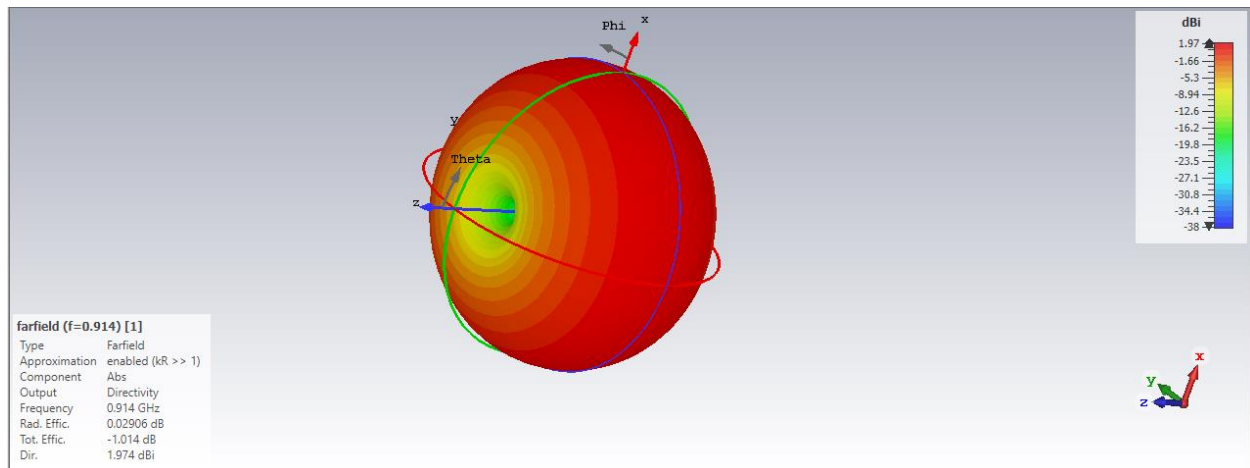


Fig: Farfield Radiation at 0.900 GHz

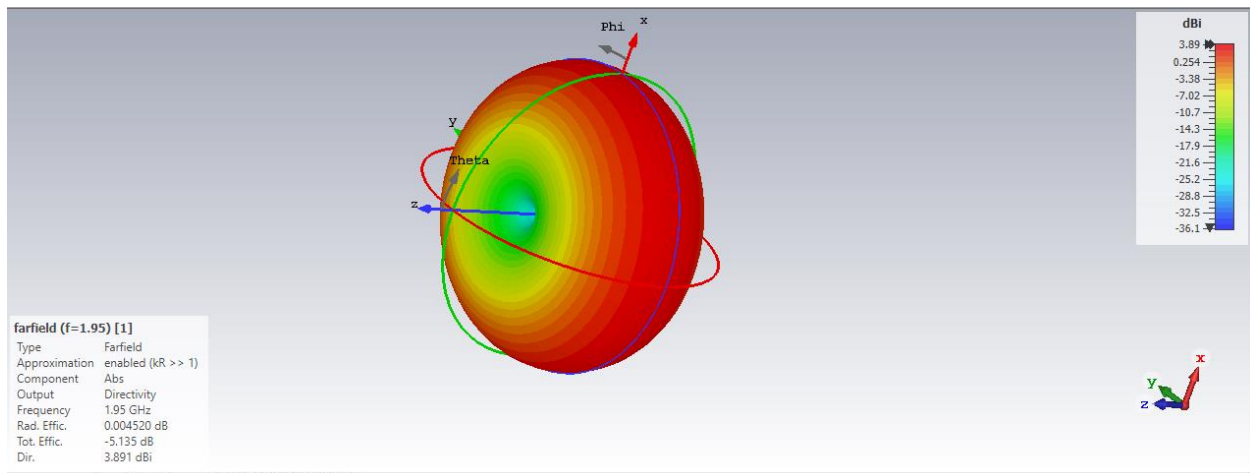


Fig: Farfield Radiation at 1.95 GHz

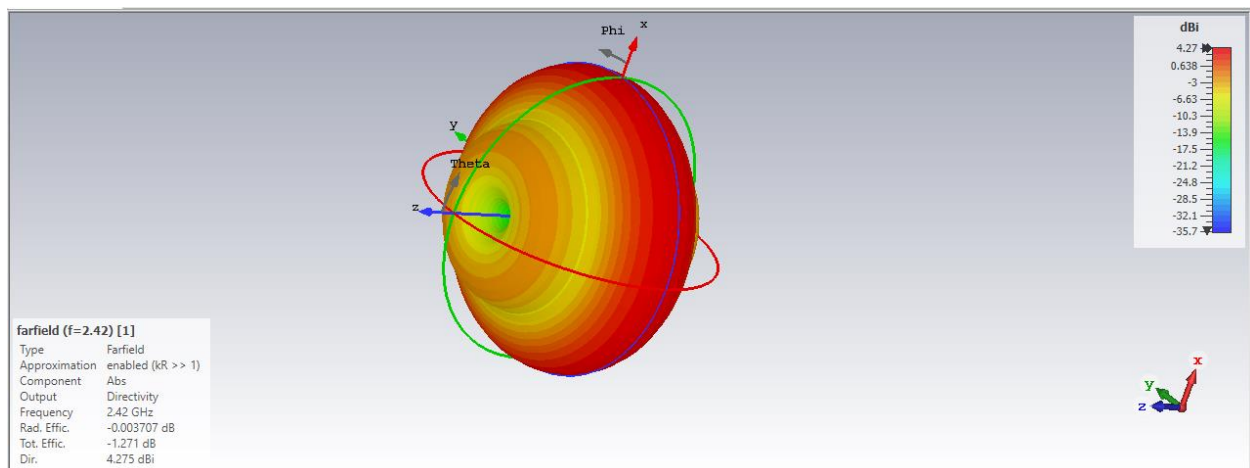


Fig: Farfield Radiation at 2.42 GHz

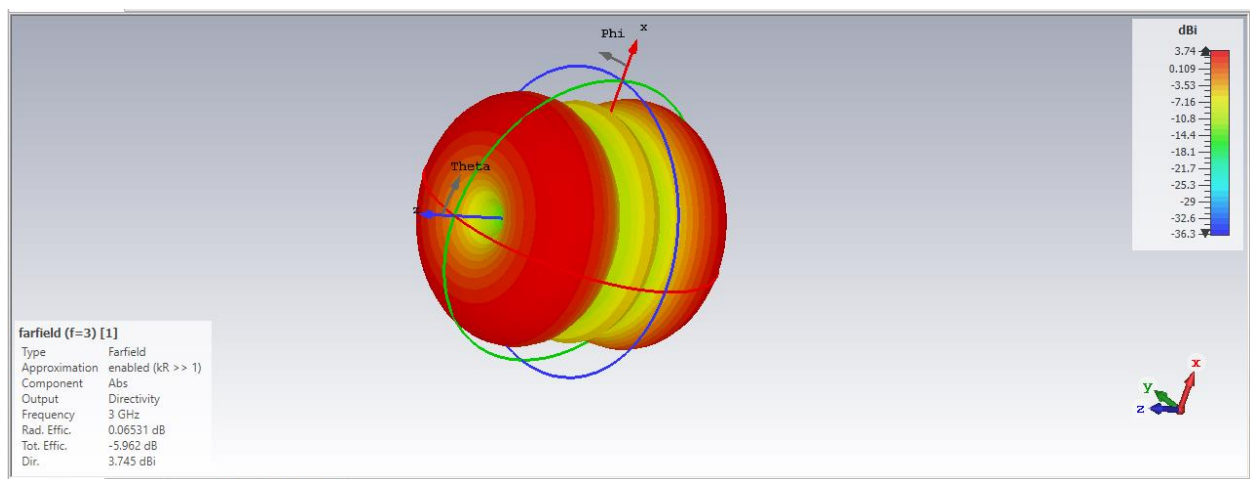


Fig: Farfield Radiation at 3 GHz

Farfield Radiation Pattern-Dipole (polar):

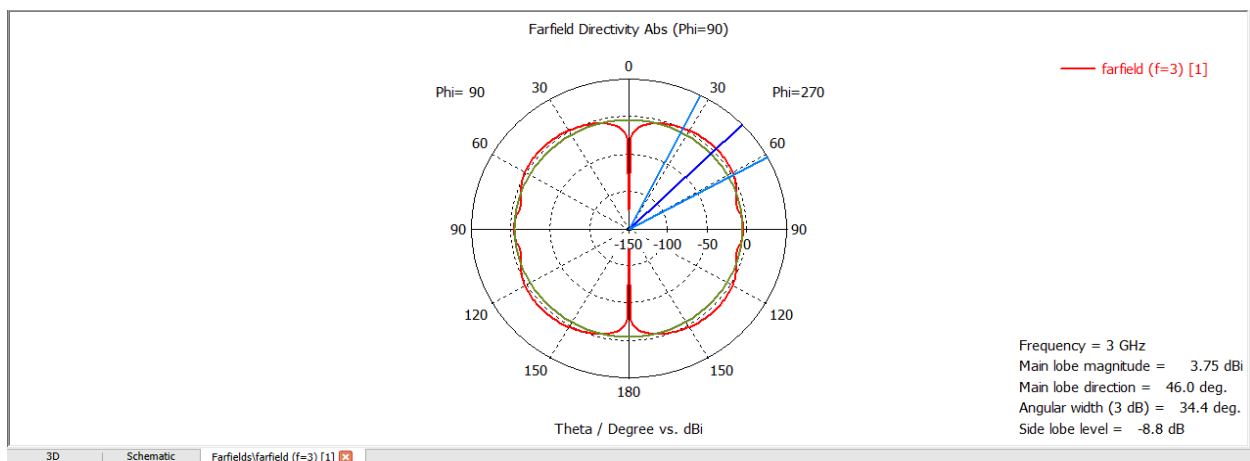


Fig: Farfield Radiation at 0.900 GHz

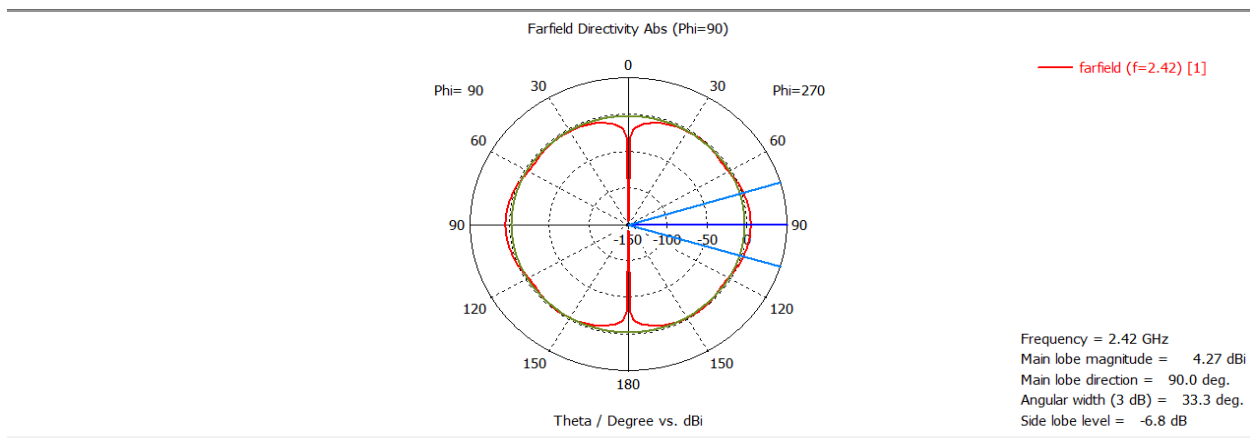


Fig: Farfield Radiation at 1.95 GHz

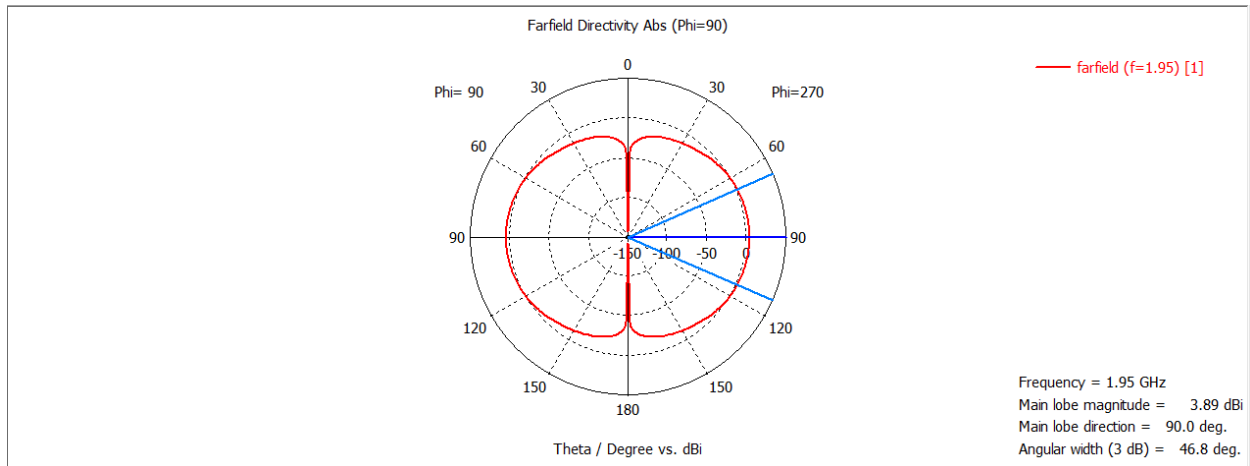


Fig: Farfield Radiation at 2.42 GHz

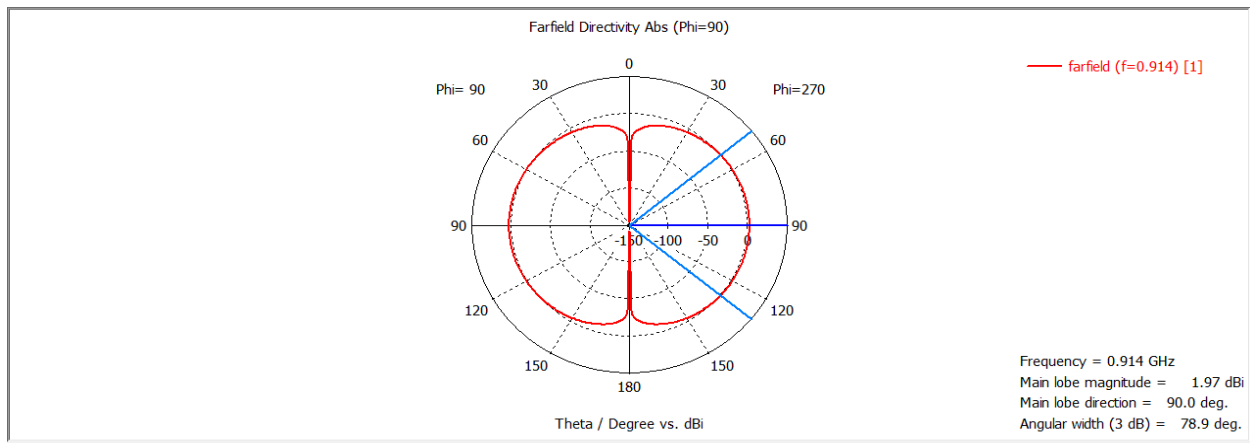


Fig: Farfield Radiation at 3 GHz

Power Radition-Dipole:

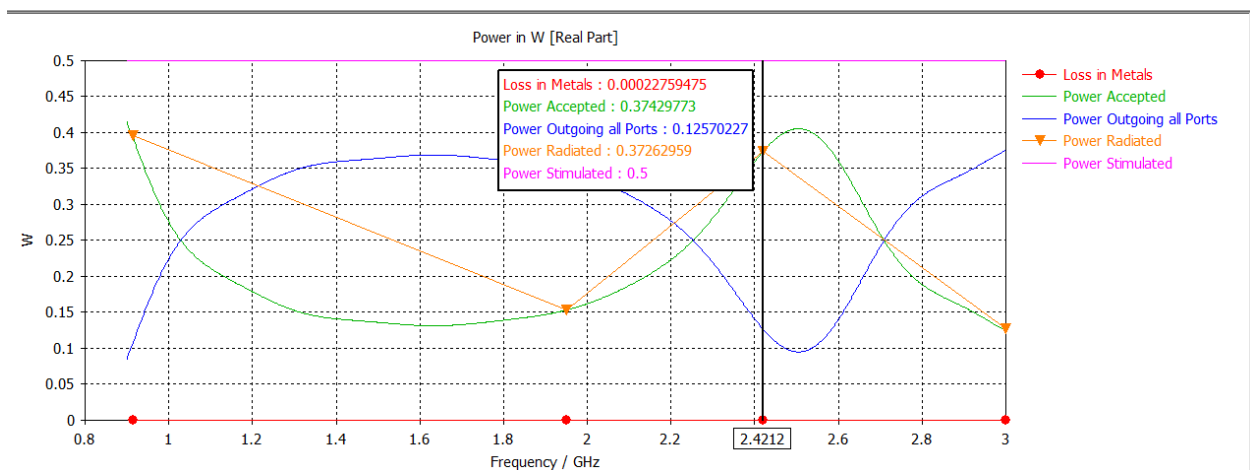


Fig: Maximum Power at Resonant Frequency 2.42 Ghz

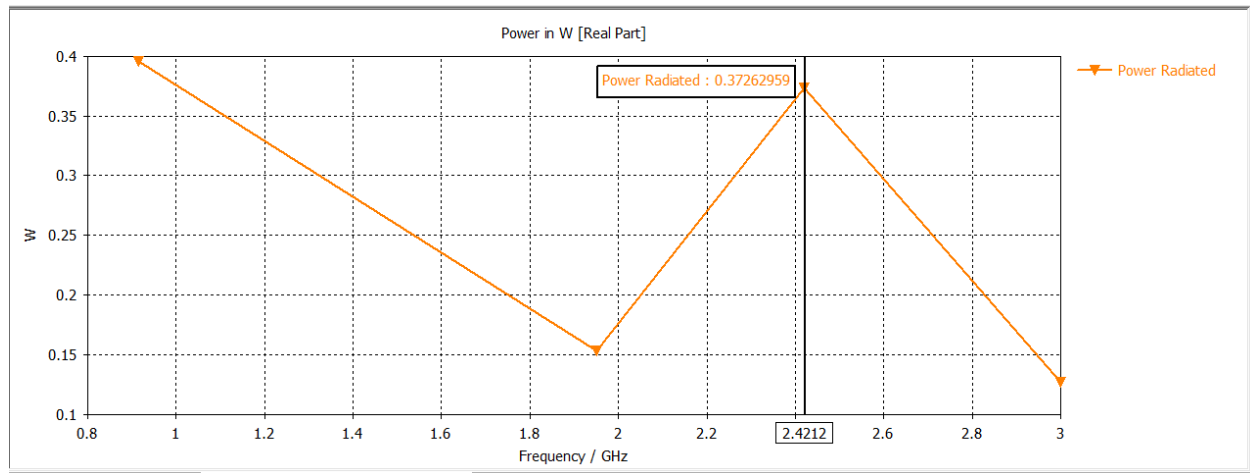


Fig: Power Radiated

Discussion

The simulation results verified the expected relationship between antenna length and resonance frequency. The parametric sweep showed that as the antenna length increases, the resonant frequency decreases. The dipole antenna exhibited a slightly broader bandwidth, whereas the monopole demonstrated higher gain due to the reflective effect of the ground plane. Minor deviations from the theoretical values were observed, primarily caused by practical considerations such as feed gap, finite ground plane dimensions, and fringing fields.

Conclusion

In this experiment, monopole and dipole antennas were successfully designed and simulated in CST Studio Suite to operate around 2.42 GHz. The dipole antenna achieved resonance at a total length of 73 mm, determined through parametric sweep analysis, while the monopole resonated with a height of 31 mm. Simulation results confirmed good radiation characteristics consistent with theoretical predictions. A minor frequency deviation observed during simulation was attributed to impedance mismatches and finite ground plane effects, which are common in practical antenna setups. Overall, both designs effectively demonstrated key antenna principles and provide a reliable foundation for real-world wireless communication applications.